

15-322 / 15-622 | FALL 2026



INTRODUCTION TO COMPUTER MUSIC

NOW
FULLY IN
PYTHON

— TURN CODE INTO SOUND

TUE / THU
11:00 AM–12:20 PM

CIC 1203

9 UNITS (15-322)
12 UNITS (15-622)

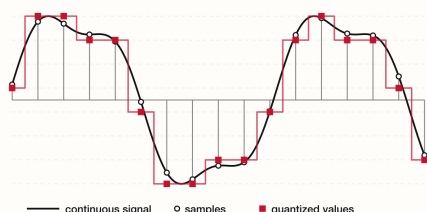
WHAT YOU'LL LEARN

DIGITAL AUDIO

SAMPLING · QUANTIZATION · ALIASING

$x(t) \rightarrow \text{sample} \rightarrow 3\text{-bit quantize}$

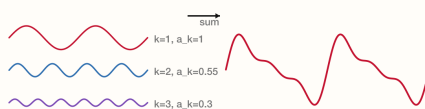
8 amplitude levels



SYNTHESIS

ADDITIVE · FM · SAMPLE-BASED · PHYSICAL MODELING

ADDITIVE: $x(t) = \sum_k a_k \sin(2\pi k f_0 t)$



FM: $y(t) = \sin(2\pi f_c t + I \sin(2\pi f_m t))$

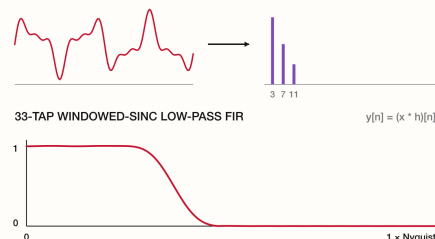


FOURIER & DSP

DFT / FFT · FILTERS · CONVOLUTION · STFT

TIME DOMAIN

DFT MAGNITUDE

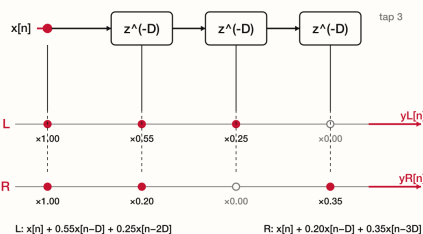


EFFECTS & SPATIALIZATION

DELAY LINES · MODULATION · PHASE VOCODING

STEREO TAPPED DELAY LINE

D = one delay interval



ALGORITHMIC COMPOSITION

MARKOV CHAINS · LPC · MIDI

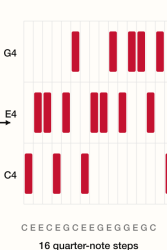
MARKOV TRANSITION MATRIX

columns = next state

rows = current

	C4	E4	G4
C4	0.10	0.60	0.30
E4	0.30	0.20	0.50
G4	0.60	0.30	0.10

SEQUENCE FROM SEED 322

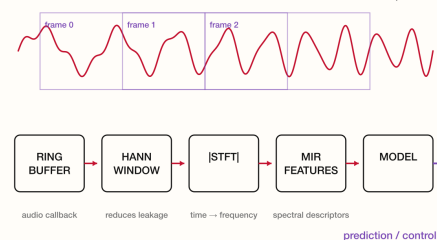


REAL-TIME AUDIO & MUSIC AI

STREAMING · MIR · MODELS

REAL-TIME FRAME PIPELINE

frame N=512 · hop H=256



9 WEEKLY ASSIGNMENTS
+ FINAL PROJECT

PREREQUISITE:
15-122 OR 15-112

MUSIC EXPERIENCE
NOT REQUIRED

INSTRUCTOR: CHRIS DONAHUE

CARNEGIE MELLON
UNIVERSITY